ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Brief Introduction to Quantum Mechanics III

I. Crystal Periodic Potential

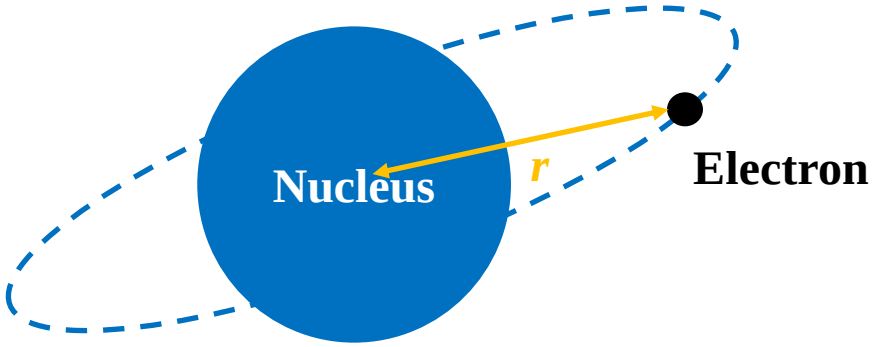
II. Bloch's Theorem & Translational Symmetry

III. Perturbation Theory

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

I. Crystal Periodic Potential

The isolated atom case



The electron feels a single Coulomb potential well. Because the electron is bound, its energy spectrum is strictly discrete

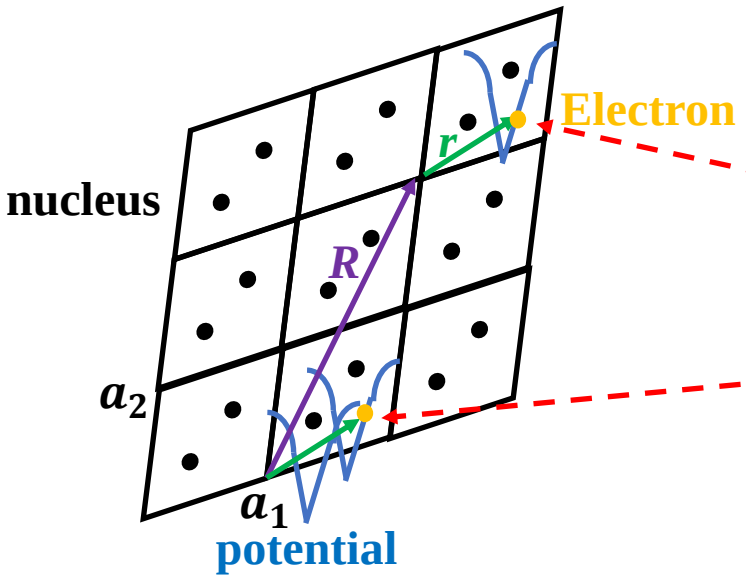
Nucleus potential $V(r) = -\frac{e^2}{r}$

Boundary condition $\Psi(r) \to 0$ **as** $r \to \infty$

Spectrum: Sharp, discrete atomic energy levels (1*s*, 2*s*, 2*p*, 3*d*, *etc*.)

I. Crystal Periodic Potential

Crystal Lattice



Infinite array of nuclei → Overlapping potential wells

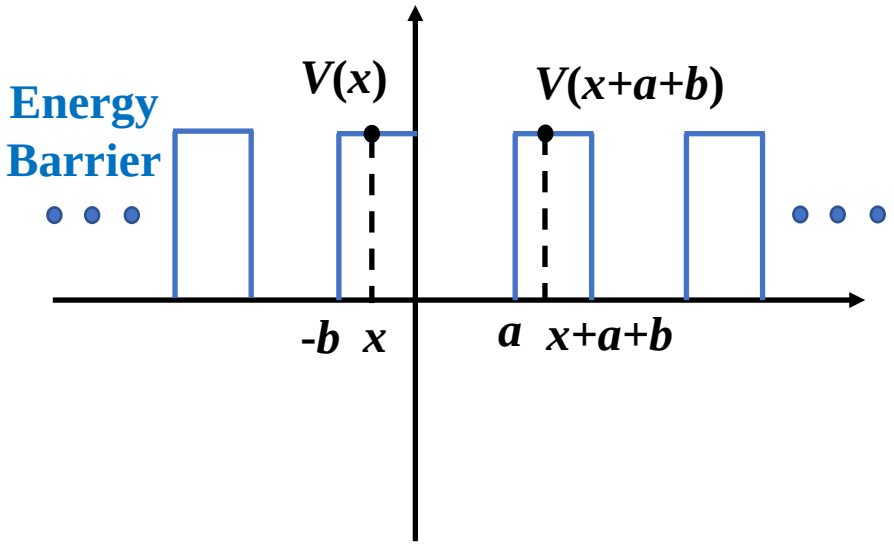
Definition of Lattice Translational Vector $R = n_1 a_1 + n_2 a_2$

Periodic potential $V(R + r) = V(r)$

In a perfect crystal, atoms are arranged periodically. Their individual potential wells overlap. The background potential $V(r)$ felt by a single electron is no longer isolated; it repeats itself perfectly across any lattice translation vector R

I. Crystal Periodic Potential

The Kronig-Penney Model: Electrons in a 1D Periodic Potential



Inside the Well (V=0): $\Psi(x) = Ae^{i\alpha x} + Be^{-i\alpha x}$

Inside the Barrier (V= V_0): $\Psi(x) = Ce^{\beta x} + De^{-\beta x}$

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\beta = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

Take the limit where the barrier becomes infinitely narrow ($b \to 0$) and infinitely high ($V_0 \to \infty$), such that the barrier area $P = lim\left(\frac{mV_0ab}{\hbar^2}\right)$ remains constant

Matching the boundary conditions yields the famous Kronig-Penney Transcendental Equation

$$P\frac{sin(\alpha a)}{\alpha a} + cos(\alpha a) = cos(ka)$$

$-b < x < 0$ — V=V_0 **Electron is forbidden by high energy barrier of width b between atoms**

$0 < x < a$ — V=0 **Electron is free inside the atom of width**

P — Barrier Strength (how hard it is for an electron to tunnel to the next atom)
a — the atomic spacing
k — the crystal momentum wavevector

I. Crystal Periodic Potential

$$P\frac{sin(\alpha a)}{\alpha a} + cos(\alpha a) = cos(ka)$$

($P \to 0$): $cos(\alpha a) = cos(ka) \to \alpha = k$ **completely continuous parabolic energy spectrum of a Free Electron**

($P \to \infty$) **electrons are trapped, recovering the discrete levels of an Isolated Particle-in-a-Box**

Real materials sit right in the middle of this tunability

The right-hand side, $cos(\alpha a)$, is trapped between -1 and 1. If you plot the left-hand side against energy, it oscillates wildly. Whenever the left-hand side overshoots 1 or drops below -1, the equation becomes mathematically impossible to solve for a real wavevector *k*.

Physically, this means if an electron tries to travel through the crystal with that specific energy, it suffers total destructive interference from the periodic lattice and gets kicked out. That is a Band Gap

II. Bloch's Theorem & Translational Symmetry

Bloch's Theorem: Eigenfunctions of a periodic Hamiltonian must take the form $\Psi_k(r) = u_k(r)e^{ikr}$

Cell-periodic function: $u_k(r) = u_k(r + R)$ **Plane wave factor**

Bloch's theorem states that any electron eigenstate in a crystal is a plane wave modulated by a periodic function
e^{ikr} **lets the electron travel through the whole crystal**
$u_k(r)$ **forces it to respect the local presence of the atomic cores**

Another form: $\Psi_k(r + R) = \Psi_k(r)e^{ikR}$

Bloch's theorem is through a phase shift
Electron jumping from one unit cell to the exact same spot in another unit cell, the wavefunction doesn't grow or decay; it simply picks up a phase factor of e^{ikR}

II. Bloch's Theorem & Translational Symmetry

Invariance of Probability Density $|\Psi_k(r+R)|^2 = \left|\Psi_k(r)e^{ikR}\right|^2 = |\Psi_k(r)|^2$

Probability of finding the electron is identical in every single unit cell
The electron belongs to the entire crystal, not to a single atom

Born-von Karman (BVK) Boundary Condition

Simulating an infinite crystal using a macro-cell of size $N_i a_i$

Boundary condition $\Psi_k(r+N_i a_i) = \Psi_k(r)$

Forces *k* to become discrete $k_i = \frac{2\pi}{a_i}\frac{m_i}{N_i}$

To handle infinity computationally, apply the Born-von Karman periodic boundary condition
Wrap the crystal into a giant loop
This mathematical trick forces the continuous wavevector *k* to become a set of densely spaced, discrete points

III. Perturbation Theory

Unperturbed Hamiltonian

Its exact eigenvalues E_n^0 and eigenstates $|n^0\rangle$ are completely known

$$H = H_0 + \lambda H'$$

A dimensionless parameter $0 \leq \lambda \leq 1$ used to track the order of correction

Perturbating Hamiltonian

A tiny, additional interaction that modifies the system

Suppose the new energy E_n and the new state $|n\rangle$ can be written as a power series. The superscripts in parentheses indicate the 'order' of the correction: '0' means unperturbed, '1' means first-order correction, and '2' means second-order correction. The goal is to find expressions for these corrections using only the known part H_0

$$E_n = E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \cdots$$

$$|n\rangle = |n^0\rangle + \lambda|n^1\rangle + \lambda^2|n^2\rangle + \cdots$$

III. Perturbation Theory

Expand $|n^1\rangle$ as a linear combination of all other unperturbed states $|n^1\rangle = \sum_{m\neq n} c_m |m^0\rangle$

$$|n^1\rangle = \sum_{m\neq n} \frac{\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0}|m^0\rangle$$

$$E_n^1 = \langle n^0|H'|n^0\rangle$$

Non-Degenerate Case

$$|n^2\rangle = \sum_{m\neq k}\sum_{k\neq n} \frac{\langle m^0|H'|k^0\rangle\langle k^0|H'|n^0\rangle}{(E_n^0 - E_m^0)(E_n^0 - E_k^0)}|m^0\rangle - \sum_{m\neq n} \frac{\langle m^0|H'|n^0\rangle\langle n^0|H'|n^0\rangle}{(E_n^0 - E_m^0)^2}|m^0\rangle - \frac{1}{2}\sum_{m\neq n} \frac{|\langle m^0|H'|n^0\rangle|^2}{(E_n^0 - E_m^0)^2}|n^0\rangle$$

$$E_n^2 = \sum_{m\neq n} \frac{|\langle m^0|H'|n^0\rangle|^2}{E_n^0 - E_m^0}$$

both a and ϕ_i can be obtained

III. Perturbation Theory

$$H_0|i^0\rangle = E^0|i^0\rangle \quad \text{for } i = 0, \ldots, f$$

Once the correct zeroth-order basis $|\alpha_k^0\rangle$ is found through diagonalization and the off-diagonal terms within the degenerate subspace is eliminated, the first-order wavefunction correction mixed in by other energy levels outside the degenerate subspace (denoted as $|m^0\rangle$ with energy $E_m^0 \neq E^0$)

Degenerate Case

$$det\begin{vmatrix} H'_{11} - E_1 & \cdots & H'_{1f} \\ \vdots & \cdots & \vdots \\ H'_{f1} & \cdots & H'_{ff} - E_1 \end{vmatrix} = 0$$

$E_i = E^0 + E_i^1$ → **Eigenstate** $|\alpha_k^0\rangle$

$$|\alpha_k^1\rangle = \sum_{E_m^0 \neq E_n^0} \frac{\langle m^0|H'|\alpha_k^0\rangle}{E^0 - E_m^0}|m^0\rangle$$

$$|\Psi_k\rangle = |\alpha_k^0\rangle + |\alpha_k^1\rangle + \cdots$$

III. Perturbation Theory

Matrix Form

$$H = \begin{vmatrix} E_1^0 + H'_{11} & V \\ 0 & V * E_2^0 + H'_{22} \end{vmatrix} = \begin{vmatrix} E_1^{diagonal} & 0 \\ 0 & V * E_2^{diagonal} \end{vmatrix}$$

The core mission of perturbation theory is to find an approximate similarity transformation—essentially a matrix rotation—that completely eliminates the off-diagonal coupling V while packing its physical effects directly into the diagonal elements

$$E_1 \approx E_1^{diagonal} + \frac{V^2}{E_1^{diagonal} - E_2^{diagonal}} = E_1^0 + H'_{11} + \frac{(H'_{12})^2}{E_1^0 - E_2^0}$$

III. Perturbation Theory

Dynamic Repulsion (Level Repulsion):

When calculating E_1, the denominator is $E_1^0 - E_2^0$. If the unperturbed energy $E_1 > E_2$ (making the denominator positive), folding the off-diagonal term into the diagonal will push E_1 even higher. Conversely, if $E_1 < E_2$ (making the denominator negative), folding the off-diagonal term in will drive E_1 even lower.

Conclusion: Once the off-diagonal coupling terms come into play, their physical effect is to push the two diagonal energy levels away from each other, widening the separation between them. In energy band theory, this is the exact matrix origin of how a band gap is forced open.

Distance Dictates Intensity (Energy Denominator Suppression):

The off-diagonal term *V* contributes to the diagonal in a squared form V^2, but its strength is scaled down by the energy difference between them. If the two energy levels are very close to each other (ΔE is small), the force of folding the off-diagonal term into the diagonal will skyrocket. If the two energy levels are far apart (ΔE is large), the off-diagonal coupling can barely make a ripple, and its contribution folded into the diagonal becomes negligible.

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

The spatial wavefunction of a hydrogen-like atom in spherical coordinates (r, θ, ϕ)

$$\varphi_{nlm_l}(r,\theta,\phi) = R_{nl}(r) \cdot Y_{lm_l}(\theta,\phi)$$

Radial wavefunction (governs the electronic shell distance and decay)

Spherical Harmonics (governs the spatial angular shape, *e.g.*, *s*, *p*, *d* shapes)

The spatial wavefunction splits beautifully into a radial part $R(r)$ and an angular part $Y(\theta, \phi)$, known as Spherical Harmonics.

For a p orbital, $l = 1$, the magnetic quantum number m_l can be 0 or ±1, which mathematically generates the peanut-like shapes of p_x, p_y and p_z orbitals in the textbook

$$p_x \qquad Y_{11} = -\sqrt{\frac{3}{8\pi}} sin\theta e^{i\phi}$$

$$p_y \qquad Y_{1-1} = \sqrt{\frac{3}{8\pi}} sin\theta e^{-i\phi}$$

$$p_z \qquad Y_{10} = \sqrt{\frac{3}{4\pi}} cos\theta$$

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

To include the electron's internal spin, multiply the spatial wave by a 2-component spinor, $|\chi\rangle$

$$|\Psi_{total}\rangle = \Psi_{nlm_l}(\boldsymbol{r}, \boldsymbol{\theta}, \boldsymbol{\phi}) \otimes |\chi_{m_s}\rangle$$

$$|\chi_{m_s}\rangle = |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}; \boldsymbol{m_s} = -\frac{1}{2}$$

$$|\chi_{m_s}\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; \boldsymbol{m_s} = \frac{1}{2}$$

The 6-fold uncoupled basis for *p* orbitals (*l* = 1) becomes a set of 2-component vector functions

$$\{|1,-1,\downarrow\rangle, |1,-1,\uparrow\rangle, |1,0,\downarrow\rangle, |1,0,\uparrow\rangle, |1,1,\downarrow\rangle, |1,1,\uparrow\rangle\}$$

Space is not enough. Electrons carry spin.

Mathematically, cross-multiply the spatial wave function with a 2-component column vector called a spinor

Spin-down is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and spin-up is $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. For the *p* orbital, this expands the Hilbert space basis into six explicit vector functions

In this basis, the wavefunction has an upper component for spin-up and a lower component for spin-down

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

Matrix Form of $L \cdot S$ and the shift operators

To calculate the SOC matrix elements, we expand the operator using ladder (shift) operators

$$L \cdot S = L_z \cdot S_z + \frac{1}{2}(L_- \cdot S_+ + L_+ \cdot S_-)$$

$$\begin{cases} L_- = L_x - iL_y\,; \quad L_+ = L_x + iL_y \\ S_- = S_x - iS_y\,; \quad S_+ = S_x + iS_y \\ L \cdot S = L_x \cdot S_x + L_y \cdot S_y + L_z \cdot S_z \end{cases}$$

where, L_-changes m_l by -1; L_+changes m_l by 1; while, $S_\pm$ flips the spin direction (↑↔↓)

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

The 1st-Order Matrix Effect

- **Diagonal elements (L_zS_z): Directly shift energies based on $m_l \cdot m_s$**
- **Off-diagonal elements ($L_{\pm}S_{\mp}$): Conjugated Coupling → It couples a state like $|1, 0, \uparrow\rangle$ to $|1, 1, \downarrow\rangle$**

The SOC operator acts on these wavefunctions. $L \cdot S$ can be rewritten using ladder operators.

The $L_z \cdot S_z$ term gives the diagonal matrix entries, while the mixing terms $L_+ \cdot S_-$ and $L_- \cdot S_+$ create off-diagonal entries. This means SOC will explicitly mix different orbital orientations with opposite spins—for instance, coupling a $p_z \uparrow$ state to a $p_x \downarrow$ state.

In the upcoming session, write down this exact 6×6 matrix and use the Adjugate Matrix method

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

$$\begin{cases} L^2|l,m_l\rangle = l(1+l)|l,m_l\rangle \\ L_z|l,m_l\rangle = m_l|l,m_l\rangle \end{cases}$$

$$\begin{cases} L^2 = L_-L_+ + L_z^2 + L_z \\ (L_-L_+ + L_z^2 + L_z)|l,l\rangle = (0 + l^2 + l)|l,l\rangle \end{cases}$$

$$\langle l,m_l|L_-L_+|l,m_l\rangle = \langle l,m_l|L^2 - L_z^2 - L_z|l,m_l\rangle = [l(1+l) - m_l^2 - m_l]\langle l,m_l\rangle|l,m_l\rangle\rangle = l(1+l) - m_l^2 - m_l$$

$$L_z(L_\pm|l,m_l\rangle) = (L_\pm L_z + [L_z, L_\pm])|l,m_l\rangle$$

$$L_z + [L_z, L_\pm] = \pm L_\pm$$

$$L_z(L_\pm|l,m_l\rangle) = L_\pm(m_l|l,m_l\rangle) \pm L_\pm|l,m_l\rangle = (m_l \pm 1)L_\pm|l,m_l\rangle$$

$L_\pm|l,m_l\rangle$ is still the eigenstate of L_z

$$L_\pm|l,m_l\rangle = N_\pm|l,m_l \pm 1\rangle$$

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

$$L_{\pm}|l,m_l\rangle = N_{\pm}|l,m_l \pm 1\rangle$$

$$\begin{cases} \langle l,m_l|L_-L_+|l,m_l\rangle = (L_+|l,m_l\rangle)^{\dagger}(L_+|l,m_l\rangle) = (N_+|l,m_l+1\rangle)^{\dagger}(N_+|l,m_l+1\rangle) = |N_+|^2\langle l,m_l+1|l,m_l+1\rangle = |N_+|^2 \\ \langle l,m_l|L_-L_+|l,m_l\rangle = l(1+l) - m_l^2 - m_l \end{cases}$$

$$|N_+|^2 = l(1+l) - m_l^2 - m_l$$

$$L_+L_- = L^2 - L_z^2 + L_z \rightarrow |N_-|^2 = l(1+l) - m_l^2 + m_l$$

$$N_{\pm} = e^{i\phi}\sqrt{l(1+l) - m_l^2 \mp m_l}$$

$$L_{\pm}|l,m_l\rangle = \sqrt{l(1+l) - m_l^2 \mp m_l}|l,m_l \pm 1\rangle$$

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

$$\begin{cases} L_{\pm}|l,m_l\rangle = \sqrt{l(1+l)-m_l^2 \mp m_l}\,|l,m_l \pm 1\rangle \\ L_z|m_l\rangle = m_l|m_l\rangle\,;\;\; S_z|m_s\rangle = m_s|m_s\rangle \end{cases}$$

Taking *p* orbital as an example

$$\begin{cases} l = 1 \begin{cases} L_+|-1\rangle = \sqrt{2}|0\rangle\,;\; L_+|0\rangle = \sqrt{2}|1\rangle\,;\; L_+|1\rangle = 0 \\ L_-|-1\rangle = 0\,;\; L_-|0\rangle = \sqrt{2}|-1\rangle\,;\; L_-|1\rangle = \sqrt{2}|0\rangle \end{cases} \\ s = 1/2: \quad S_-|\uparrow\rangle = |\downarrow\rangle\,;\; S_+|\downarrow\rangle = |\uparrow\rangle \end{cases}$$

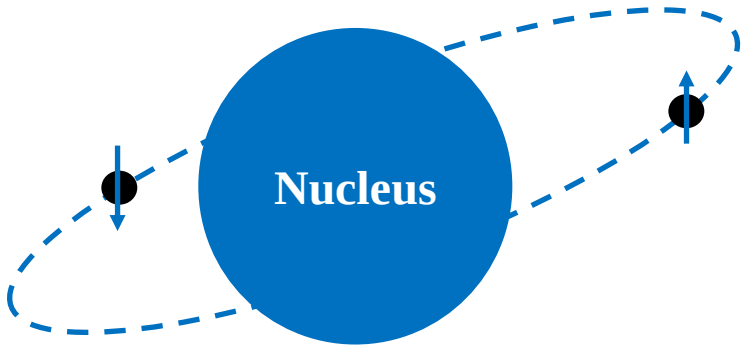
$\langle m_l, m_s|L\cdot S|m_l, m_s\rangle =$

	$\lvert 1,\uparrow\rangle$	$\lvert -1,\downarrow\rangle$	$\lvert 1,\downarrow\rangle$	$\lvert 0,\uparrow\rangle$	$\lvert 0,\downarrow\rangle$	$\lvert -1,\uparrow\rangle$
$\lvert 1,\uparrow\rangle$	1/2	0	0	0	0	0
$\lvert -1,\downarrow\rangle$	0	1/2	0	0	0	0
$\lvert 1,\downarrow\rangle$	0	0	-1/2	$1/\sqrt{2}$	0	0
$\lvert 0,\uparrow\rangle$	0	0	$1/\sqrt{2}$	0	0	0
$\lvert 0,\downarrow\rangle$	0	0	0	0	0	$1/\sqrt{2}$
$\lvert -1,\uparrow\rangle$	0	0	0	0	$1/\sqrt{2}$	-1/2

IV. Mathematical Representation of Atomic Orbitals & Spin Orbit Coupling (SOC)

$$\langle m_l, m_s | L \cdot S | m_l, m_s \rangle =$$

	$\|1,\uparrow\rangle$	$\|-1,\downarrow\rangle$	$\|1,\downarrow\rangle$	$\|0,\uparrow\rangle$	$\|0,\downarrow\rangle$	$\|-1,\uparrow\rangle$
$\|1,\uparrow\rangle$	1/2	0	0	0	0	0
$\|-1,\downarrow\rangle$	0	1/2	0	0	0	0
$\|1,\downarrow\rangle$	0	0	-1/2	$1/\sqrt{2}$	0	0
$\|0,\uparrow\rangle$	0	0	$1/\sqrt{2}$	0	0	0
$\|0,\downarrow\rangle$	0	0	0	0	0	$1/\sqrt{2}$
$\|-1,\uparrow\rangle$	0	0	0	0	$1/\sqrt{2}$	-1/2



p Shell Orbital Electron

Through Diagonalization

- A 4-fold degenerate state with E = 1/2 (Total angular momentum j = 3/2)
- A 2-fold degenerate state with E = -1 (Total angular momentum j = 1/2)



That's all for this session

Thank you for your attention and great participation